

Who Moved the Mode?

Modal Self-Energy Feedback Cores and Exact Project Spectral Responsibility in Inverter-Dominated Power Systems

Jorge L. Mayorga Taborda

Abstract—Inverter-dominated grids are commonly interpreted through bus-level network strength, device participation factors, or full-model eigenanalysis. These views identify weak locations or participating states, but they do not directly answer three increasingly important planning questions: which closed interaction creates a global oscillatory mode, which finite combination of projects moves that mode, and which smallest intervention can remove the harmful mechanism without undoing physically beneficial upgrades. This paper develops a hierarchical answer. First, exact Schur elimination maps hidden controller and subnetwork dynamics into a frequency-dependent self-energy acting on retained ports. A residue expansion decomposes each subgraph self-energy mode by mode, yielding a modal interaction graph whose strongly connected components localize determinant-level feedback cores. Second, finite line, BESS/GFM, and controller projects are represented as low-rank perturbations. Their project Green matrix gives an exact perturbation determinant in project space. Contour spectral moments then recover the finite displacement of an isolated critical pole exactly; finite Möbius inversion decomposes that displacement into coalition spectral dividends, while closed project-walk traces provide a representation-invariant cycle interpretation. Existing IEEE 39-bus evidence supplies a demanding mechanism-level falsification set: branch 19–20 improves algebraic connectivity, a bus-34 impedance proxy, and losses yet reduces controller-family damping from 2.89% to 0.186%; a passive hidden cycle gain of 0.00596 becomes 1.013 after controller dressing and is suppressed by controller retuning; all 46 single branch actions and all 1035 pairs are stable at a common $1.35\times$ reinforcement while a pre-identified three-action portfolio is unstable; and 46,000 nonlinear trajectories preserve the controller-aware action ordering while physical λ_2 does not. The result is a framework that moves from participation to closed mechanism and from local sensitivity to exact finite-project spectral responsibility.

Index Terms—inverter-based resources, small-signal stability, self-energy, modal interaction, feedback cycle, Schur complement, Green function, perturbation determinant, spectral moments, project portfolio, dynamic security, grid-forming converter.

I. INTRODUCTION

LOW-INERTIA power systems increasingly combine synchronous generators, grid-following (GFL) and grid-forming (GFM) converters, BESS, HVDC, fast inner controls, and network reinforcements. The engineering problem is no longer only whether the base operating point is stable.

J. L. Mayorga Taborda is with Universidad de los Andes, Bogota, Colombia. This manuscript distinguishes exact algebraic results from mechanism-level IEEE 39-bus evidence. Full manufacturer-grade DAE/EMT replication is not claimed.

Planners must repeatedly answer whether a transmission project, converter siting decision, controller retuning, or synchronous retirement is dynamically beneficial when combined with other projects.

Existing tools resolve important pieces of this problem. Impedance methods characterize converter-grid return ratios [2], [3]; participation factors and extended modal-impedance methods identify influential states or control loops [13]; graph-theoretic approaches connect topology and stability for specific inverter-grid model classes [10]; sensitivity methods map line or operating-point perturbations to eigenvalues [11], [12]; and recent interconnection-aware reduction frameworks preserve subsystem interactions in large inverter grids [14]. These advances leave a narrower gap. A global mode may be a collective object that does not belong to one device, and a harmful finite project portfolio may arise through interaction even when every lower-order subset is individually acceptable.

The question pursued here is therefore:

Can a global inverter-driven mode be reduced to a small closed feedback core, and can the actual finite displacement of that mode be assigned exactly to the project coalitions that created it?

The proposed answer combines two representations. In *mechanism space*, Schur self-energy and modal residues convert hidden dynamics into effective mode-to-mode feedback. In *project space*, a low-dimensional Green operator converts finite interventions into a perturbation determinant whose contour moments track the critical spectrum. The mathematical ingredients—Schur complements, component-mode ideas, Birman–Schwinger-type operators, the argument principle, and Möbius inversion—are established. The contribution is their specialization into a controller-aware power-system calculus that connects *what created the mode* to *which project moved it*.

The main contributions are:

- C1. Hierarchical modal self-energy.** Exact boundary reduction is combined with a residue expansion that assigns the external self-energy of a subgraph to its internal modes.
- C2. Modal feedback-core factorization.** A graph whose vertices are retained dynamical modes is constructed; determinant-level interaction factorizes over its strongly connected components (SCCs), localizing emergent feedback modes to small cores.

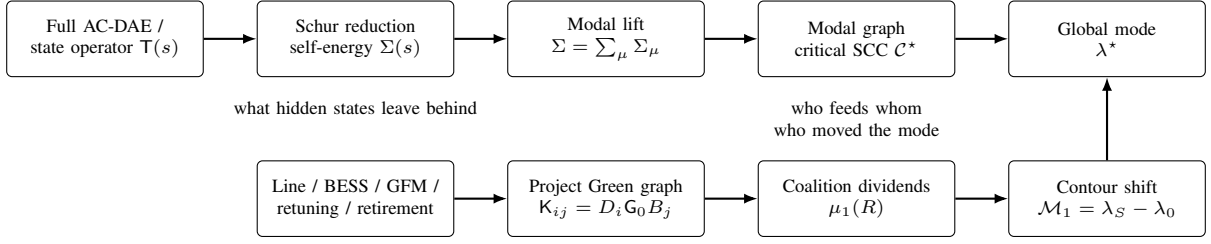


Fig. 1. Two complementary representations of the same dynamic event. The upper path localizes the physical/controller mechanism that sustains an emergent mode. The lower path assigns the finite displacement of that mode to project coalitions.

C3. Exact finite-project spectral responsibility. Low-rank project perturbations produce a project Green matrix. Contour spectral moments return the exact finite displacement of an isolated critical pole, and Möbius inversion decomposes that displacement into intrinsic coalition dividends.

C4. Power-system falsification evidence. The theory is anchored to an IEEE 39-bus mechanism campaign containing a controller-mediated reinforcement reversal, hidden-cycle amplification, all-action nonlinear transfer, high-resolution co-design, and a subset-stable three-action portfolio instability.

Fig. 1 summarizes the framework. The paper intentionally does not claim that Schur reduction, modal analysis, or cooperative-game algebra are new. It claims a new power-system use: a hierarchical path from hidden controller dynamics to minimal feedback cores and from finite projects to exact spectral responsibility.

II. FROM FULL DYNAMICS TO HIERARCHICAL SELF-ENERGY

A. Exact retained-space operator

After regular algebraic elimination of a linearized DAE, partition the dynamic states into retained interface/network coordinates x_r and internal device/controller states x_h ,

$$\dot{x} = Ax, \quad A = \begin{bmatrix} A_{rr} & A_{rh} \\ A_{hr} & A_{hh} \end{bmatrix}. \quad (1)$$

With $T_r = sI - A_{rr}$ and $T_h = sI - A_{hh}$, exact elimination gives

$$\boxed{T_{\text{eff}}(s) = T_r(s) - \Sigma(s)}, \quad \Sigma(s) = A_{rh}T_h(s)^{-1}A_{hr}. \quad (2)$$

The characteristic determinant factorizes as

$$\det(sI - A) = \det T_h(s) \det T_{\text{eff}}(s). \quad (3)$$

Equation (2) is the controller/subnetwork self-energy: hidden states do not disappear; they re-enter the retained model as frequency-dependent effective coupling.

For an arbitrary subgraph with boundary variables b and internal variables i ,

$$T_k(s) = \begin{bmatrix} T_{bb,k} & T_{bi,k} \\ T_{ib,k} & T_{ii,k} \end{bmatrix}, \quad (4)$$

its exact boundary operator is

$$\boxed{\Lambda_k(s) = T_{bb,k} - T_{bi,k}T_{ii,k}^{-1}T_{ib,k}}. \quad (5)$$

This is the dynamic analogue of a boundary/Thevenin equivalent, but it is matrix-valued and frequency dependent.

Theorem 1 (Hierarchical Schur equivalence): Assume the internal blocks of a collection of subgraphs are invertible and the subgraphs interconnect only through retained boundary ports. Eliminating each subgraph internally and then assembling the boundary operators gives the same retained operator as assembling first and eliminating the union of all internal states.

Proof: Schur complementation is associative for nested elimination under the stated invertibility conditions. Equivalently, block Gaussian elimination performed locally or globally produces the same Schur complement on the common retained coordinates. ■

The theorem is standard matrix algebra; its role here is to justify an exact graph-of-dynamic-equivalents representation before any modal truncation.

B. Self-energy decomposes by internal mode

Assume temporarily that the internal state matrix $A_{ii,k}$ is diagonalizable with right/left eigenvectors $v_{k\mu}, w_{k\mu}$ normalized by $w_{k\mu}^H v_{k\nu} = \delta_{\mu\nu}$. Then

$$(sI - A_{ii,k})^{-1} = \sum_{\mu} \frac{v_{k\mu} w_{k\mu}^H}{s - \lambda_{k\mu}}, \quad (6)$$

and therefore

$$\boxed{\Sigma_k(s) = \sum_{\mu} \frac{r_{k\mu} \ell_{k\mu}^H}{s - \lambda_{k\mu}}}, \quad (7)$$

where $r_{k\mu} = T_{bi,k} v_{k\mu}$ and $\ell_{k\mu}^H = w_{k\mu}^H T_{ib,k}$.

Proposition 1 (Modal self-energy expansion): The external dynamic effect of a subgraph is the sum of its modal residue contributions in (7). Consequently, an effective retained edge can be attributed to internal modes rather than only to a physical device or bus.

This produces a useful diagnostic distinction: participation factors identify which states belong to a global mode, whereas (7) identifies which internal modes modify the boundary dynamics seen by the rest of the grid.

III. MODAL INTERACTION GRAPH AND FEEDBACK CORES

Retain a set of local modes $a = (k, \mu)$ from the subgraphs. Let λ_a denote their uncoupled poles and let $\kappa_{ab} = w_a^H C v_b$

denote the inter-subgraph modal coupling induced by the boundary interconnection C . Define

$$D_M(s) = \text{diag}\left(\frac{1}{s - \lambda_1}, \dots, \frac{1}{s - \lambda_m}\right), \quad (8)$$

$$R_M(s) = D_M(s)K_M.$$

with $[K_M]_{ab} = \kappa_{ab}$. Interaction-induced poles satisfy

$$\det[I - R_M(s)] = 0. \quad (9)$$

Self terms may be absorbed into a diagonal block D ; after the corresponding normalization let Q_M contain only cross-mode interactions.

Theorem 2 (Modal feedback-core factorization): Permute Q_M into block upper-triangular form according to its SCCs Q_c . Then

$$\det(I - Q_M) = \prod_{c \in \text{SCC}} \det(I - Q_c). \quad (10)$$

If the modal interaction graph is acyclic, $\det(I - Q_M) = 1$.

Proof: The SCC condensation graph is acyclic, hence a topological ordering makes Q_M block upper triangular. The determinant of $I - Q_M$ is the product of determinants of its diagonal SCC blocks. If every SCC is a singleton and the diagonal has been removed, Q_M is strictly triangular. ■

This yields the central mechanism statement. Feed-forward modal couplings can transmit disturbances, but determinant-level emergent feedback resides in SCCs. Define a *minimal modal feedback core* C^* for a critical mode λ^* as the smallest retained SCC/subgraph whose characteristic root reproduces λ^* within a prescribed tolerance. The concept is analogous in spirit to component-mode synthesis in structural dynamics [15], but here the retained object is a closed interaction among heterogeneous controller/network modes rather than a vibration substructure alone.

Fig. 2 shows the mechanism already present in the IEEE 39-bus benchmark. The retained physical ports {16, 33, 34} contain no direct physical edges, yet Schur condensation creates a frequency-dependent effective graph. The passive cycle remains negligible while controller dressing drives the 33–34 cycle close to unity.

IV. EXACT FINITE-PROJECT SPECTRAL RESPONSIBILITY

A. Project Green matrix

Let project i —a branch reinforcement, BESS/GFM installation, controller retuning, damping change, or retirement action—produce a finite-rank perturbation

$$\Delta T_i(s) = B_i(s)D_i(s). \quad (11)$$

For coalition S , stack the factors into B_S, D_S and write

$$T_S(s) = T_0(s) + B_S(s)D_S(s), \quad G_0 = T_0^{-1}. \quad (12)$$

Define the project Green matrix

$$K_S(s) = D_S(s)G_0(s)B_S(s), \quad (13)$$

whose (i, j) block is $D_i G_0 B_j$. The determinant lemma gives

$$\Delta_S(s) \equiv \frac{\det T_S(s)}{\det T_0(s)} = \det[I + K_S(s)]. \quad (14)$$

Thus the full-grid propagation remains in G_0 , while portfolio interaction is represented in a space whose dimension is the sum of project ranks.

The factorization of a project is not unique. If $B'_i = B_i S_i$ and $D'_i = S_i^{-1} D_i$ for nonsingular S_i , then $K'_S = S^{-1} K_S S$ with block-diagonal S . Hence Δ_S , its zeros, and spectra/traces of closed project walks are representation invariant.

B. Contour moments track the actual pole

Let Γ be a positively oriented contour on which T_0, T_S , and Δ_S are nonsingular. Define

$$\mathcal{M}_k(S; \Gamma) = \frac{1}{2\pi j} \oint_{\Gamma} s^k \frac{\Delta'_S(s)}{\Delta_S(s)} ds. \quad (15)$$

Theorem 3 (Exact spectral-moment shift): If Δ_S is meromorphic inside Γ and nonzero on Γ , then

$$\mathcal{M}_k(S; \Gamma) = \sum_{\lambda \in Z_S} m_{\lambda} \lambda^k - \sum_{\nu \in P_S} m_{\nu} \nu^k, \quad (16)$$

where Z_S and P_S are the zeros and poles of Δ_S inside Γ . In particular, if Γ encloses exactly one base mode λ_0 and its perturbed continuation λ_S ,

$$\mathcal{M}_1(S; \Gamma) = \lambda_S - \lambda_0. \quad (17)$$

Proof: Apply the weighted argument principle to the meromorphic perturbation determinant Δ_S . ■ Unlike a first-order eigenvalue sensitivity, (17) is a finite-action statement.

C. Coalition dividends and project cycles

Define the finite Möbius transform

$$\mu_k(R; \Gamma) = \sum_{S \subseteq R} (-1)^{|R|-|S|} \mathcal{M}_k(S; \Gamma). \quad (18)$$

Then

$$\lambda_A - \lambda_0 = \sum_{\emptyset \neq R \subseteq A} \mu_1(R; \Gamma) \quad (19)$$

whenever the isolated-mode condition of (17) holds. $\text{Re } \mu_1(R)$ is the intrinsic coalition contribution to decay-rate motion and $\text{Im } \mu_1(R)$ the contribution to modal-frequency motion.

For scalar/rank-one project blocks, split $K = K_d + K_o$ and let $Q = (I + K_d)^{-1} K_o$. In any region where the logarithmic series converges,

$$\log \det(I + Q) = \sum_{p=2}^{\infty} \frac{(-1)^{p+1}}{p} \text{tr}(Q^p). \quad (20)$$

The traces expand into closed project walks. For a pair $\{a, b\}$ the unique two-project feedback factor is $q_{ab} q_{ba}$; for block projects, $\text{tr}(Q_{ab} Q_{bc} \cdots Q_{za})$ is invariant under internal project basis changes. Therefore the coalition spectral dividend is an exact responsibility object, while the cycle expansion provides a mechanism interpretation when its convergence certificate is satisfied.

IEEE 39-bus numerical mechanism: a physical line action amplifies a hidden controller-mediated cycle

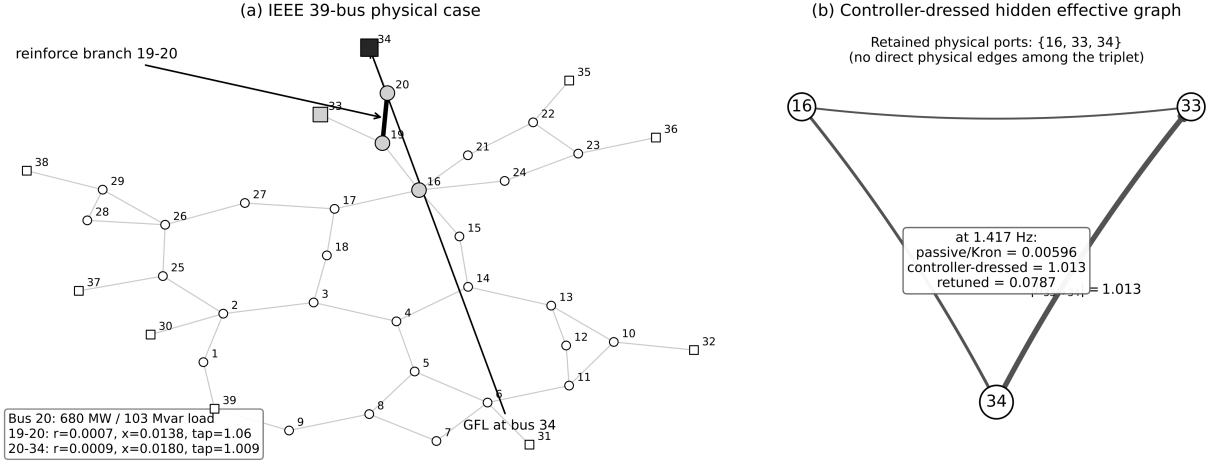


Fig. 2. IEEE 39-bus mechanism used as the power-system anchor. Branch 19–20 reinforcement improves the physical network, while Schur/controller dressing produces a hidden effective cycle on retained ports $\{16, 33, 34\}$. At 1.417 Hz the passive/Kron 33–34 cycle gain is 0.00596, the controller-dressed value is 1.013, and controller retuning reduces it to 0.0787 without removing the reinforcement.

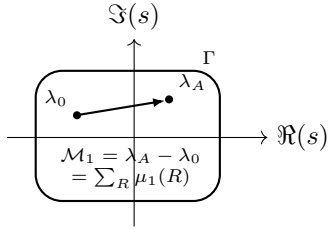


Fig. 3. Exact contour spectral responsibility. The perturbation determinant tracks mode count and finite pole displacement; Möbius inversion assigns that displacement to project coalitions.

V. IEEE 39-BUS EVIDENCE AND PLANNING CONSEQUENCES

A. Controller-mediated reinforcement reversal

The physical benchmark is the standard MATPOWER IEEE 39-bus case. A transparent GFL surrogate is connected at bus 34 and fixed before screening all branch actions. The central branch action reinforces 19–20 by $1.8\times$. Algebraic connectivity increases by 0.694%, the bus-34 impedance proxy decreases by 4.60%, and AC losses decrease by 0.227%. Nevertheless, the hidden feedback margin falls by more than 90%, and controller-family damping falls from 2.89% to 0.186%. Near a $1.925\times$ action the mode crosses the imaginary axis. Retuning only K_ω from 1.65 to 2.20 at the same reinforced topology moves the critical pole to $-0.502 + j9.282$ rad/s and restores 5.40% damping.

These numbers matter because they separate three interpretations. The physical graph becomes stronger; raw controller self-energy alone is insufficient; the failure appears only after controller dressing is propagated through the network resolvent and returned through the feedback loop. Across all 46 branch actions, physical λ_2 , a Fiedler-only projection, and raw Σ

mixing are weakly associated with the hidden margin, while the full-loop spectral radius is strongly associated with it.

B. Subset-stable portfolio instability

The portfolio result is more consequential for planning. All $\binom{46}{2} = 1035$ two-branch portfolios are evaluated at common action levels. At $1.35\times$, all 46 single actions and all 1035 pairs remain stable, but the pre-identified three-action portfolio

$$A = 19\text{--}20, \quad B = 19\text{--}33, \quad C = 16\text{--}19 \quad (21)$$

is unstable, even though physical λ_2 is higher than in the base case. Dense subset continuations give

$$\alpha_A^* = 1.9201, \quad \alpha_B^* = 2.4921, \quad (22)$$

$$\alpha_{AB}^* = 1.3829, \quad \alpha_{AC}^* = 1.5566, \quad (23)$$

$$\alpha_{BC}^* = 2.3063, \quad \boxed{\alpha_{ABC}^* = 1.3375}. \quad (24)$$

The result must not be misread as “the third-order Möbius term is destabilizing”; in the present benchmark it is slightly stabilizing at $1.35\times$. The stronger statement is structural: cumulative lower-order interactions compress the safe portfolio radius so that a higher-order coalition crosses first. This is precisely the setting in which exact spectral dividends are useful: they separate intrinsic single, pair, and higher-order responsibility for the actual mode motion instead of inferring causation from marginal ranks.

C. Nonlinear transfer and uncertainty

The linear controller-aware ranking is challenged with a common nonlinear stochastic protocol rather than action-specific tuning. With 1000 trajectories for each of the 46 branch actions (46,000 trajectories total), the full-loop spectral radius preserves the nonlinear controller-RMS ranking with Spearman $\rho = 0.984$, ROC AUC 0.990 for the worst quartile,

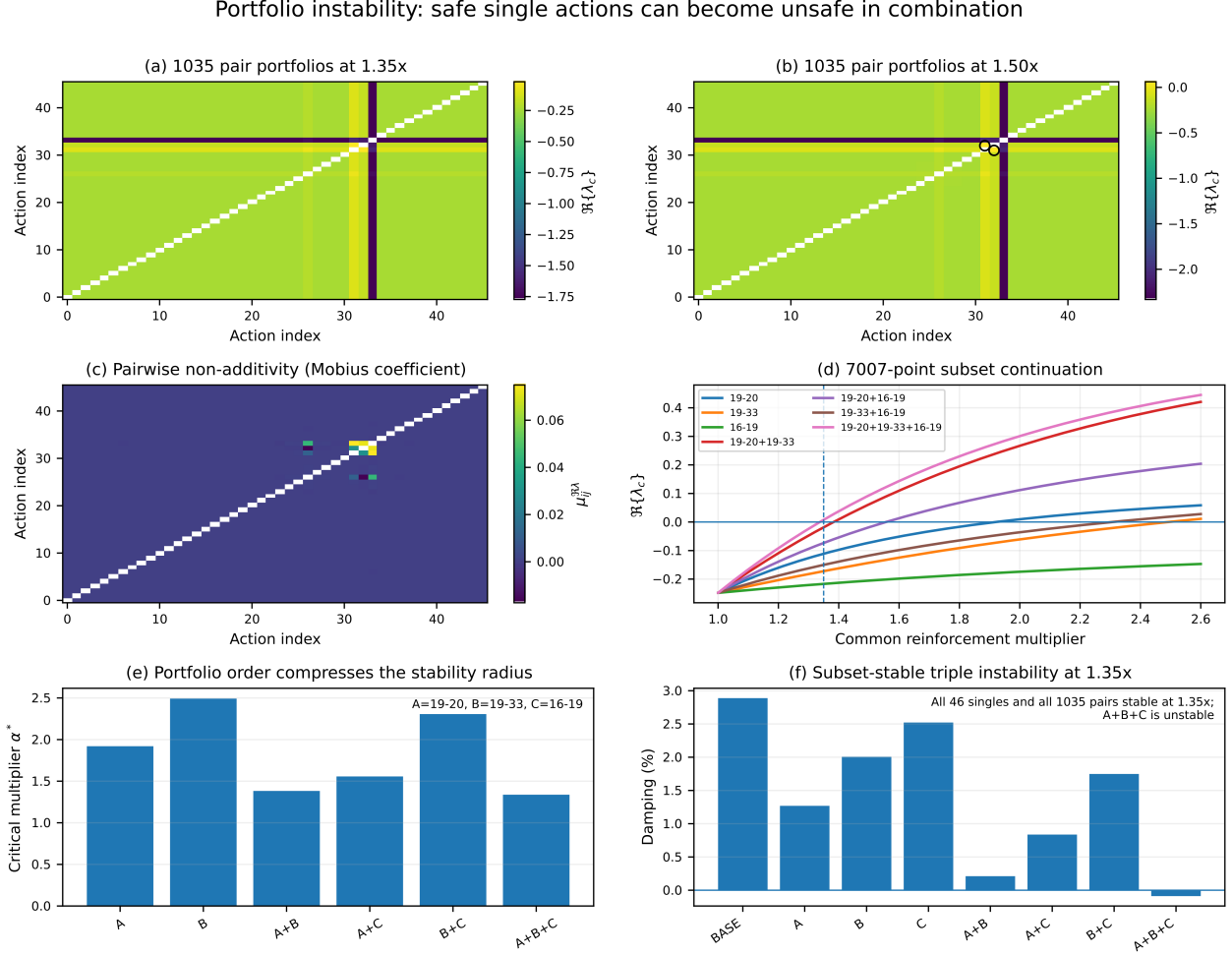


Fig. 4. Portfolio non-composability in the IEEE 39-bus benchmark. At $1.35\times$ every single action and all 1035 pairs are stable, while the three-action portfolio $A + B + C$ is unstable. The 7007-point subset continuation shows how interaction order compresses the safe action radius.

and exact top-10 overlap 1.00. Hidden-margin collapse gives $\rho = 0.842$ and top-10 overlap 0.80, whereas physical λ_2 gives $\rho = 0.047$ and zero top-10 overlap. In a separate 30,000-trajectory star-case ensemble, reinforcement increases median controller RMS from 0.1158 to 0.2013 (+73.8%); retuning reduces it to 0.05985, 48.3% below base. Finally, among 6199 base-stable controller-parameter draws in a 10,000-draw Monte Carlo, damping degrades in every draw after reinforcement, 19.1% cross instability, and the same fixed controller-only retuning restores stability in all 6199.

VI. WHAT THE FRAMEWORK ADDS TO EXISTING POWER-SYSTEM TOOLS

Table I gives the intended novelty boundary. The proposed framework does not replace full eigensolvers, impedance analysis, or model reduction. It adds two answers that those tools do not provide directly: a closed minimal interaction core and an exact finite-project decomposition of the critical mode displacement.

The connection to structural dynamics is useful but should not be overclaimed. Component mode synthesis reduces large

structures to interface coordinates plus selected internal modes [15], [16]; recent inverter-grid reduction similarly preserves subsystem interconnection structure [14]. The present use is different: modal residues of self-energy are lifted into a directed feedback graph, and the objective is causal stability localization rather than only reduced-order fidelity.

Likewise, the project Green matrix is mathematically close to a finite-dimensional Birman–Schwinger operator [17]. The publishable power-system contribution is the interpretation of its auxiliary space as a portfolio of heterogeneous grid projects and the use of its perturbation determinant to assign finite critical-mode motion. Shapley allocation, if desired for economic cost sharing, is downstream of the physical Möbius dividends and is not required for the theory.

VII. INDUSTRIAL USE AND LIMITATIONS

The framework suggests a practical multi-fidelity workflow. Conventional AC constraints first remove physically unattractive projects. A controller-aware operator is then built from the full or trusted reduced dynamic model. Hidden self-energy/feedback cores identify which devices or regions require

Massive Monte Carlo validation: 46,000 all-action trajectories + 30,000 star-case trajectories

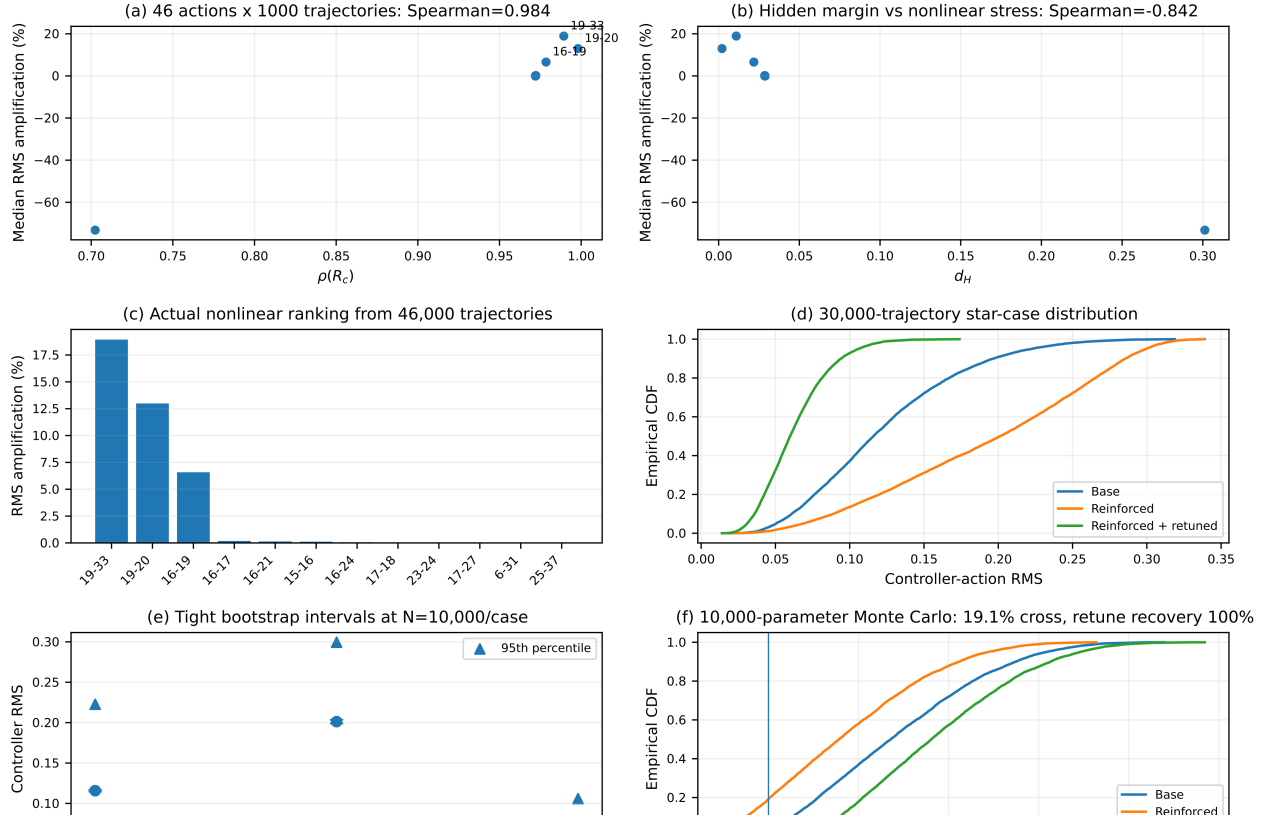


Fig. 5. Large-sample nonlinear and uncertainty evidence. The controller-aware full-loop and hidden-margin screens preserve nonlinear stress ordering across 46 actions, while the physical graph does not. The same controller-only remedy also survives a 10,000-draw parameter uncertainty test.

TABLE I
DIAGNOSTIC INFORMATION PROVIDED BY REPRESENTATIVE METHODS

Method	Weak/critical location	Participating state/device	Directed coupling	Closed feedback core	Finite project attribution
Physical SCR / λ_2 / Fiedler	✓	—	—	—	—
Eigenvalue participation/sensitivity	partial	✓	—	—	local/infinitesimal
Impedance / modal impedance	✓	✓	partial	partial	—
Interconnection-aware MOR	partial	partial	partial	—	—
Proposed hierarchy	✓	✓	✓	✓	exact finite shift

joint treatment. Candidate projects are lifted into the project Green graph. Acyclic project structures compose at determinant level; SCCs are screened jointly. For flagged portfolios, contour spectral responsibility reports which coalition moved the mode and by how much. Remediation can then target the smallest controllable edge or controller parameter in the critical core while retaining useful transmission or BESS investments.

The present manuscript deliberately keeps its strongest engineering claim narrower than its mathematical one. The IEEE 39-bus evidence is an internally consistent and heavily falsified mechanism study, not a manufacturer-grade multi-IBR validation. The GFL model was chosen to expose the mechanism transparently, and the nonlinear trajectories share that surrogate model class. Therefore the exact algebraic identities and the contour responsibility calculus are ready independently of model fidelity, but industrial universality

requires independent DAE/EMT replication with documented SG/GFL/GFM models. The frozen dangerous-action ordering (19–20, 19–33, 16–19), the subset-stable portfolio, and the controller-only remedy provide prespecified tests for that next validation step.

A second limitation is modal truncation. The hierarchical Schur reduction is exact before truncation; a modal graph becomes approximate when only selected residues are retained. Near hidden feedback closure, even small discarded residues can be amplified by the return-difference inverse. A decision-preserving truncation certificate—rather than an eigenvalue-error certificate alone—is therefore an important extension.

VIII. CONCLUSION

This paper develops a unified answer to the question “who moved the mode?” in inverter-dominated grids. Hidden

controller and subnetwork states are first converted into exact self-energy acting on retained ports. Modal residues of that self-energy produce a graph of interacting dynamical mechanisms, and determinant factorization localizes collective feedback to SCC cores. Finite grid projects are then represented in a second, low-dimensional project graph. Its perturbation determinant admits contour moments that recover the finite critical-pole displacement exactly, while Möbius inversion assigns that displacement to project coalitions.

The IEEE 39-bus benchmark shows why these objects are useful. A physically beneficial branch reinforcement can amplify a controller-mediated hidden cycle and nearly eliminate damping; controller-only retuning repairs the mode without removing the line. More importantly, safe single actions and safe pairs do not guarantee a safe multi-action portfolio: at $1.35\times$ all 46 single actions and all 1035 pairs are stable, yet a pre-identified triple is unstable. Large-sample nonlinear tests further show that the controller-aware loop geometry, not the physical Laplacian alone, predicts the stress ordering of branch actions. The resulting planning message is specific: global inverter-driven modes should be diagnosed as closed interaction cores, and finite project decisions should be attributed to the actual spectral motion they produce rather than to isolated marginal scores.

ACKNOWLEDGMENT

OpenAI ChatGPT was used to assist manuscript organization, language editing, and LaTeX drafting. All technical claims, equations, numerical results, and cited sources remain the responsibility of the author.

REFERENCES

- [1] P. Kundur, *Power System Stability and Control*. New York, NY, USA: McGraw-Hill, 1994.
- [2] J. Sun, "Impedance-based stability criterion for grid-connected inverters," *IEEE Trans. Power Electron.*, vol. 26, no. 11, pp. 3075–3078, 2011.
- [3] M. Lu, Y. Yang, B. Johnson, and F. Blaabjerg, "An interaction-admittance model for multi-inverter grid-connected systems," *IEEE Trans. Power Electron.*, vol. 34, no. 8, pp. 7542–7557, 2019.
- [4] X. Lin *et al.*, "Improving small-signal stability of grid-connected inverter under weak grid by decoupling phase-lock loop and grid impedance," *IEEE Trans. Ind. Electron.*, vol. 69, no. 7, pp. 7040–7053, 2022.
- [5] F. Dörfler and F. Bullo, "Kron reduction of graphs with applications to electrical networks," *IEEE Trans. Circuits Syst. I*, vol. 60, no. 1, pp. 150–163, 2013.
- [6] D. I. Shuman *et al.*, "The emerging field of signal processing on graphs," *IEEE Signal Process. Mag.*, vol. 30, no. 3, pp. 83–98, 2013.
- [7] R. D. Zimmerman, C. E. Murillo-Sánchez, and R. J. Thomas, "MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education," *IEEE Trans. Power Syst.*, vol. 26, no. 1, pp. 12–19, 2011.
- [8] H. Cui, F. Li, and K. Tomsovic, "Hybrid symbolic-numeric framework for power system modeling and analysis," *IEEE Trans. Power Syst.*, vol. 36, no. 2, pp. 1373–1384, 2021.
- [9] L. Ding *et al.*, "Holistic small-signal stability analysis for large-scale inverter-intensive power systems with coupled and full-order dynamics from control systems and power networks," *IEEE Trans. Ind. Appl.*, vol. 61, no. 1, pp. 1131–1147, 2025.
- [10] I. Bačić *et al.*, "A graph theoretic view on small signal stability of inverter-based power grids," arXiv:2607.08260, 2026.
- [11] T. Joswig-Jones, S. Dong, J. Tan, and B. Zhang, "Sensitivity-based system strength assessment: Mapping power flow and network topology perturbations to system eigenvalues," arXiv:2607.28764, 2026.
- [12] J. Kook and J.-W. Park, "An interaction sensitivity framework in perturbation space for the converter-driven stability of power systems," arXiv:2607.25334, 2026.
- [13] L. Zheng, J. Zheng, L. Chen, and C. Liu, "Extended impedance modal analysis of internal dynamics in grid-following inverters," arXiv:2503.00797, 2025.
- [14] A. Sedlak, P. Serna-Torre, and P. Hidalgo-Gonzalez, "Model reduction of electromagnetic transient dynamics for inverter-based grids: An interconnected systems framework," *Electr. Power Syst. Res.*, Art. no. 113671, online 2026, doi: 10.1016/j.epsr.2026.113671.
- [15] R. R. Craig, Jr. and M. C. C. Bampton, "Coupling of substructures for dynamic analyses," *AIAA J.*, vol. 6, no. 7, pp. 1313–1319, 1968.
- [16] R. R. Craig, Jr. and C.-J. Chang, "A review of substructure coupling methods for dynamic analysis," NASA CR-2781, 1977.
- [17] J. Behrndt, A. F. M. ter Elst, and F. Gesztesy, "The generalized Birman–Schwinger principle," *Trans. Amer. Math. Soc.*, vol. 375, no. 2, pp. 799–845, 2022.
- [18] F. Gesztesy, H. Holden, and R. Nichols, "On factorizations of analytic operator-valued functions and eigenvalue multiplicity questions," *Integral Equations Operator Theory*, vol. 82, pp. 61–94, 2015.
- [19] J. C. Harsanyi, "A bargaining model for the cooperative n -person game," in *Contributions to the Theory of Games IV*. Princeton, NJ, USA: Princeton Univ. Press, 1959, pp. 325–355.
- [20] K. N. Hasan, R. Preece, and J. V. Milanovic, "Application of game theoretic approaches for identification of critical parameters affecting power system small-disturbance stability," *Int. J. Electr. Power Energy Syst.*, vol. 97, pp. 344–352, 2018.
- [21] D. Witthaut and M. Timme, "Braess's paradox in oscillator networks, desynchronization and power outage," *New J. Phys.*, vol. 14, Art. no. 083036, 2012.